

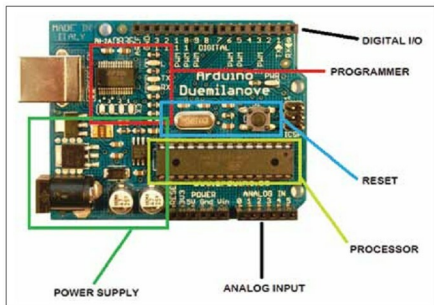


Fig. 2: The Arduino hardware

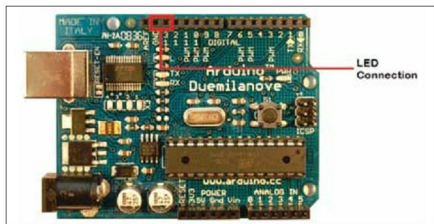


Fig. 3: Connecting the LED

external power is an option). You can connect an AC-DC adaptor (~12V, with centre pin positive). The board can take up to 25V DC. Also, you will require an FTDI cable (serial-to-USB) if you do not have a COM port in your computer.

Duemilanove users can just plug-and-play after installing the drivers. The FTDI USB drivers are with the Arduino IDE in the 'drivers' folder.

Let's get our hands wet

Here, the program 'Blink' is used to explain, using the Duemilanove. Even if the serial COM port version is used there will hardly be any difference per

se. The Arduino hardware is shown in Fig. 2.

The digital input/output (I/O) pins are where you can connect output devices like motors, LCDs, touchscreens and sensors for input. The analogue input pins are where input sensors and analogue input devices like potentiometers (pots) can be connected. The processor—ATmega—is the brain of the entire operation.

Download the Arduino IDE from the website and install it in your system if you haven't already done so. Extract it. The Arduino can be programmed from Windows, Mac and Linux. Start the IDE and go to

File→Examples→Digital→Blink. The code for blinking LED is listed below:

```
int ledPin = 13; // LED connected
to digital pin 13

// The setup() method runs once, when
the sketch starts

void setup() {
  // initialize the digital pin as
  an output:
  pinMode(ledPin, OUTPUT);
}

// the loop() method runs over and
over again,
// as long as the Arduino has power

void loop()
{
  digitalWrite(ledPin, HIGH); //
  set the LED on
  delay(1000); //
  wait for a second
  digitalWrite(ledPin, LOW); //
  set the LED off
  delay(1000); //
  wait for a second
}
```

Arduino source code contains the following sections:

1. The setup(), which runs only once (automatically) when the program starts running.
2. The loop(), which is an infinite loop that is triggered once setup() completes.
3. Other user-defined functions.

In any program, the setup() and loop() are mandatory. Any extra libraries used will have to be imported using the 'import' statement before the setup().

Coming back to the LED blinking example, the first line sets an integer variable with the value of the pin at which we have connected the LED. The 'pinMode' function tells the ATmega whether the pin 13 ('ledPin' variable's value) has an input or an output device connected to it. The first argument is the pin number. The LED is an output component, hence pinMode's second argument would be output.